JAVA PROGRAMMING PROJECT PRESENTATION

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Scholarship Eligibility Verification and Student Shortlisting System

Course: Java Programming Lab / Mini Project

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INTRODUCTION

This project is a Scholarship Eligibility Verification and Student Shortlisting System developed using Java. It helps to:

- Verify whether students are eligible for scholarships.
- Check students' academic marks and family income.
- Shortlist eligible students based on predefined criteria.
- Display scholarship eligibility and selection status.

The system reduces manual work and improves the accuracy and efficiency of scholarship selection.

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OBJECTIVE

The main objectives of the project are:

- To design a scholarship eligibility verification system.
- To verify student details and academic performance.
- To check family income against scholarship requirements.
- To shortlist eligible students automatically.
- To apply object-oriented programming concepts in Java.
- To improve the speed and accuracy of scholarship processing.

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PROBLEM STATEMENT

Manual scholarship selection systems face several problems:
• Time-consuming verification of student applications.
• Errors in checking marks and family income.
• Difficulty in identifying eligible students.
• Delays in preparing the shortlisted student list.
• Possibility of inconsistent application of eligibility rules.
This project addresses these problems by automating eligibility verification and student shortlisting using Java.

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SYSTEM REQUIREMENTS

Hardware Requirements
- Processor: Intel i3 or higher.
- RAM: Minimum 4 GB.
- Storage: 500 MB of free space.

Software Requirements
- Operating System: Windows or Linux.
- Java Development Kit (JDK).
- IDE: Eclipse, IntelliJ IDEA, or Visual Studio Code.
- Java compiler and runtime environment.

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SYSTEM DESIGN

The system consists of the following modules:

- Student Details Module – Collects student information.
- Academic Verification Module – Checks marks and academic performance.
- Income Verification Module – Verifies family income.
- Eligibility Checking Module – Determines whether students meet scholarship criteria.
- Student Shortlisting Module – Generates the list of eligible students.

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MODULE DESCRIPTION

1. Student Details Module
Stores student name, student ID, course, academic marks, and family income.
2. Academic Verification Module
Checks whether the student’s marks meet the minimum academic requirements.
3. Income Verification Module
Compares family income with the specified scholarship income limit.
4. Eligibility Module
Uses conditional statements to determine whether the student satisfies all required conditions.
5. Student Shortlisting Module
Displays eligible students and their selection status based on defined criteria.

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ALGORITHM / FLOWCHART

START
↓
Enter student details
↓
Verify academic marks and family income
↓
Check eligibility criteria
↓
Eligible? — Yes → Add student to shortlist
No → Display not eligible status
↓
Display results
↓
STOP

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IMPLEMENTATION

The project uses the following Java programming concepts:
• Classes and Objects: Represent student records and their details.
• Conditional Statements: Use if-else to verify eligibility.
• Loops: Process multiple student applications.
• Methods: Separate verification, eligibility checking, and shortlisting.
• Arrays or Collections: Store and process student information.
• Input and Output: Accept student details and display results.
Main features include academic eligibility checking, income verification, and automatic generation of the eligible student list.

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OUTPUT SCREENS

SCHOLARSHIP ELIGIBILITY VERIFICATION

Enter Student Name: Arun
Enter Student ID: 101
Enter Academic Marks (%): 85
Enter Family Income: 120000

Eligibility Status: ELIGIBLE
Student Added to Shortlist Successfully.

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RESULTS & DISCUSSION

- The system verifies eligibility according to predefined criteria.
- Academic marks and family income are checked automatically.
- Eligible students are added to the shortlist.
- Manual verification effort is reduced.
- Results are displayed quickly and consistently.
- The system demonstrates how Java can automate scholarship application processing.

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CONCLUSION & FUTURE ENHANCEMENT

Conclusion
- Designed the proposed scholarship eligibility verification system.
- Automated student eligibility checking and shortlisting.
- Applied Java object-oriented programming concepts.
- Improved speed and consistency of application processing.

Future Enhancements
- Develop a GUI using Java Swing or JavaFX.
- Connect the application to a MySQL database.
- Add student registration and login.
- Generate reports and export shortlisted details to PDF or Excel.
- Add an administrator module to manage scholarship criteria.

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REFERENCES / ACKNOWLEDGEMENT

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THANK YOU

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